

Medical Transcriptions Classification

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Qualitative Analysis

Background / Motivation

Hospitals handle many thousands of clinical notes and discharge summaries daily. Senior physicians now spend a substantial amount of time reading them manually to sort patients by medical specialty and identify critical symptoms. Reading these notes manually is both labor-intensive and error-prone, creating a bottleneck in patient routing for areas of high volume.

Most current clinical NLP systems are based on a simple keyword match and do not account for medical shorthand, abbreviations, and negatives (for example, treating "no fever" as equivalent to "fever"). Therefore, there is a strong need for an automated pipeline to process raw clinical transcriptions through cleaning and comprehension to produce an accurate classification of the patient(s) being reported on their own. This project fills this need by developing an end-to-end NLP pipeline using real clinical data from the MTSamples dataset.

Objectives

1. Clean raw medical transcriptions by removing header noise, expanding abbreviations, and standardizing text through lemmatization.
2. Extract key medical entities (symptoms, body parts, severity descriptors) using POS tagging.
3. Detect negation in clinical sentences to distinguish present symptoms from denied or absent ones.
4. Classify each transcription into the correct medical specialty (Surgery, Cardiovascular/Pulmonary, Orthopedic, Radiology) using TF-IDF and machine learning.
5. Evaluate model performance and analyse misclassification patterns to understand clinical vocabulary overlap.

Methods

The MTSamples dataset (4,999 transcriptions from 40 different areas) will be analyzed. The categories of 'Consult-History & Phy' & 'SOAP/Chart/Progress Notes' were excluded in order to focus solely on the specialties with the greatest total number of transcriptions. The four specialties with the most number of records were Surgery (1088), Cardiovascular/Pulmonary (371), Orthopedic (355), Radiology (273). If there are more than 273 records in any of the areas selected for analysis, all but 273 records will be undersampled in order to create a balanced dataset of 1,092 records for the purpose of conducting an equal comparison of the analytic results.

Phase 1 (Pre processing): The use of a regular expression based function has been used to remove structured headers (Sample Name, ID, Date, Description) from transcriptions within the dataset. A dictionary of 22 medical abbreviations (e.g., pt -> patient, hx -> history, sob -> shortness of breath) was applied to eliminate abbreviations within transcriptions. The module built into spaCy (en_core_web_sm) will perform tokenization & lemmatization of all tokens; this will standardize all forms of inflected words that are included within the final dataset. A custom created sample of 40+

generic terms (Patient, Procedure, Left, Right, Normal) from Medical stop-words was utilized to eliminate these universal words from affecting the specialty specific analysis.

Phase 2 (Entity Extraction): The POS tagging capabilities of spaCy were used to extract all nouns (NOUN, PROPN) as symptoms or body parts, and adjectives (ADJ) as descriptors of severity; Tokens that are less than three letters and belonged to MEDICAL_STOPWORDS were filtered out of the dataset. Frequency analysis of the filtered dataset revealed the five most frequent specialty specific nouns for each of the classes selected for analysis. Word clouds were created using only the noun tokens to visually verify the differences in vocabulary utilized between each of the two selected specialties for comparison (Cardiovascular/Pulmonary orthope) to support the analysis of frequency data.

Phase 3 was to leverage spaCy's dependency parser to establish syntactic relationships. For the purpose of negation detection, the function `detect_negation()` was used to locate negation terms (for example: no, not, denies, without, absent, and also negative/unremarkable) by traversing the dependency tree structure using three different paths (ancestor nodes, child nodes, and sibling nodes). After the completion of this analysis the identifiers of every noun entity within the report were tagged as either Negated or Present.

Phase 4 was to use TF-IDF (`max_features=15,000`; `ngram_range=(1,2)`; `sublinear_tf=True`; and `stop_words='english'`) to vectorize the raw transcription text. Stratified 80/20 train/test splits were applied across the data sets that were used to train both the Multinomial Naive Bayes (`alpha=0.1`) and Logistic Regression (`C=1`, lbfgs solver) models. Evaluation of model performance was through the use of accuracy, precision, recall, F1-score, and confusion matrix metrics.

Results

Phase 1 reduced the average text length by removing headers and eliminating noise from each note or report by 15 to 20 percent. It was able to correctly expand all 22 shorthand forms, including the correct expansion of slash pattern abbreviations (i.e. r/o and c/o) through using look ahead patterns. The lemmatization process was utilized to combine like words by taking fractured/fractures and combining them into the word fracture; combining complained/complaining into the word complain.

Phase 2 vocabulary analysis determined distinct noun distributions based on each specialty performed. The specialty of surgery contained predominately operative terminology, such as incisions, sutures, wounds, abdomen, and gallbladder. The specialty of cardiovascular/pulmonary contained vascular anatomy terms, such as artery, valve, coronary, and stenosis. The specialty of orthopedics contained musculoskeletal terminology, such as knee, fracture, joint, and bone. The specialty of radiology contained terminology related to imaging, such as image, evidence, mass, and exam. Word clouds would visually demonstrate and confirm that there were very few similarities between the top 20 terms from cardiovascular/pulmonary and orthopedics (no/negligible overlap).

Phase 3 negation detection correctly identified negated entities in 5 out of the 7 sentences used during the validation of negation detection. Examples are as follows: Patient denies chest pain = pain, negated (trigger = denies); no fever or chills = fever, negated (trigger = no); heart sounds normal without murmur = murmur, negated (trigger = without); patient has severe headache = headache, present.

Phase 4 classification accuracy results were 70.32% (Naive Bayes) and 72.60% (Logistic regression/best model). Both models achieved balanced per-class evaluation (F1-scores 0.69-0.75) indicating that no singular class was systematically neglected. The confusion matrix displays the classification as Surgery → Orthopedic (13 samples, 23.6%); Radiology → Cardiovascular (9 samples, 16.4%).

Conclusion

This project demonstrates that an NLP pipeline for clinical report triage leveraging clinical text cleaning, POS-based entity extraction, dependency-based negation detection, and TF-IDF classification can automate triaging with 72.60% accuracy with balanced clinical specialties. The NLP pipeline produced usable results that are applicable to real-world clinical events, such as medical abbreviations, header noise, and negation issues. All misclassification events are clinically explainable since a shared operative vocabulary exists between the disciplines of Surgery and Orthopedic. Subsequent enhancements could include utilizing fine-tuned transformer models (including BioBERT and ClinicalBERT), developing named entities for medical concepts, and implementing confidence threshold limits to identify uncertain predictions for human review.

Quantitative Analysis

2.1 Dataset and Preprocessing

Source: MTSamples dataset (Kaggle) – 4,999 transcriptions across 40 specialties. Administrative categories excluded before selection.

Rank	Medical Specialty	Original Records	After Balancing
1	Surgery	1,088	273
2	Cardiovascular / Pulmonary	371	273
3	Orthopedic	355	273
4	Radiology	273	273
	Total	2,087	1,092

Preprocessing pipeline applied in sequence:

- **Regex header removal:**

```
r'Sample Name:.*?\n' → removes sample name lines
r'ID:.*?\n' → removes ID fields
r'Date:.*?\n' → removes date fields
r'Description:.*?\n' → removes description headers
r'\(Medical Transcription Sample Report\)' → removes boilerplate tag
```

- **Abbreviation expansion:**

Abbreviation	Expansion	Abbreviation	Expansion
pt	patient	sob	shortness of breath
hx	history	cp	chest pain
dx	diagnosis	bp	blood pressure
tx	treatment	hr	heart rate
fx	fracture	r/o	rule out
w/o	without	c/o	complains of
h/o	history of	s/p	status post
y/o	year old	wbc	white blood cell
rx	prescription	rbc	red blood cell
sx	symptoms	w/	with

- Lemmatization via spaCy en_core_web_sm + custom MEDICAL_STOPWORDS (40+ generic terms)

2.2 Model Design / Algorithm Implementation

Two models were trained on TF-IDF features:

Parameter	Naive Bayes	Logistic Regression
Algorithm	Multinomial NB	Logistic Regression
Key Hyperparam	alpha = 0.1	C = 1, solver = lbfgs
Multi-class	Native	Multinomial (auto)
Class weight	None (data balanced)	None (data balanced)
Software	scikit-learn 1.x	scikit-learn 1.x

TF-IDF Vectorizer configuration:

Parameter	Value	Reason
max_features	15,000	Capture sufficient vocabulary
ngram_range	(1, 2)	Unigrams + bigrams for phrase context
min_df	2	Ignore ultra-rare terms
max_df	0.85	Ignore near-universal terms
sublinear_tf	True	Log-normalization for long documents
stop_words	'english'	Built-in stopwords removal

2.3 Training and Evaluation Setup

Data split: 80% train / 20% test, stratified by specialty (random_state=42).

Split	Samples	Per Class
Training	873	~218
Test	219	~55
Total	1,092	273

Evaluation metrics used: Accuracy, Precision (macro), Recall (macro), F1-Score (macro), Confusion Matrix.

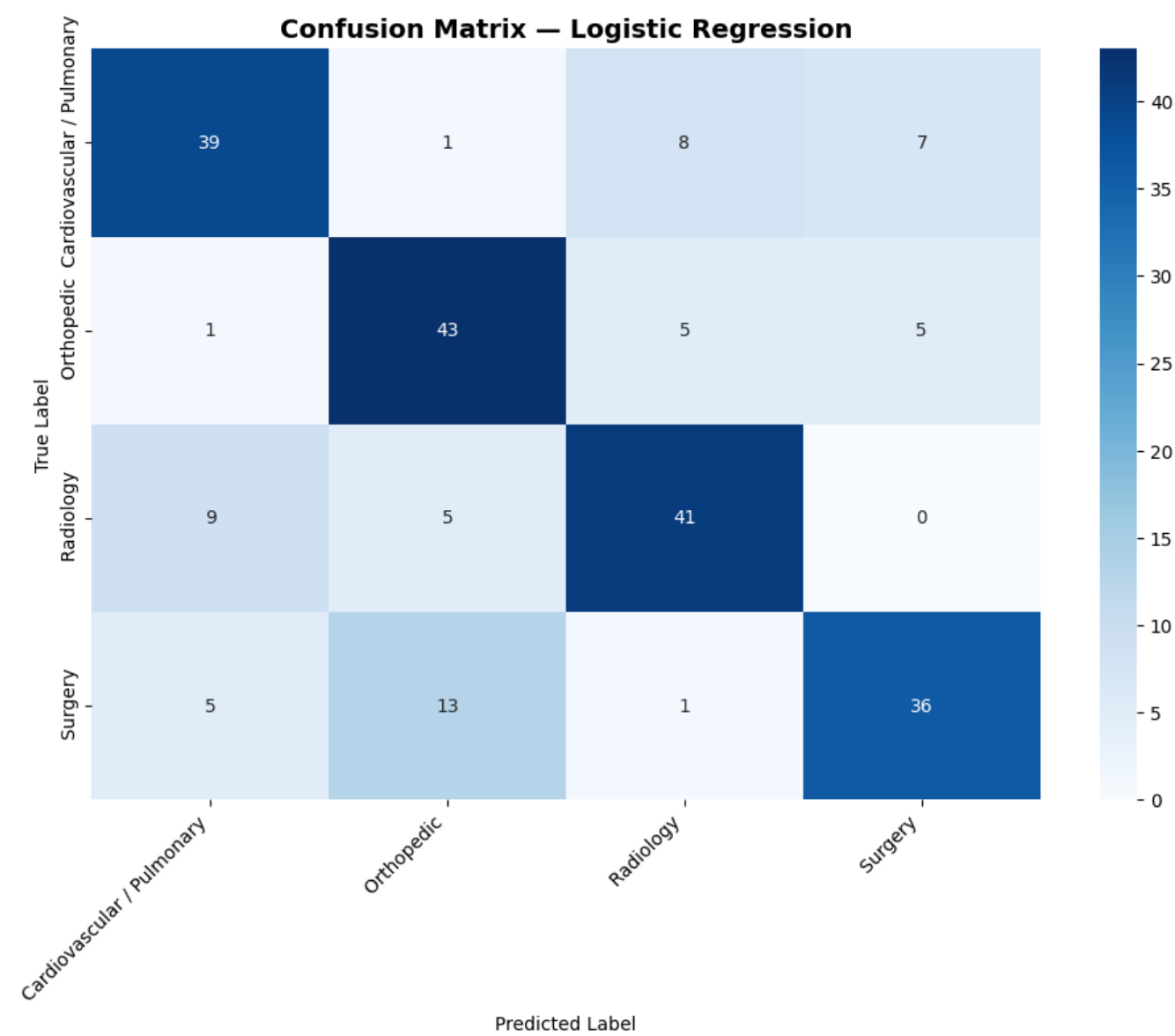
2.4 Experimental Results

Model	Accuracy	Notes
Multinomial Naive Bayes	70.32%	alpha=0.1; fast, strong baseline
Logistic Regression	72.60%	C=1, lbfgs solver; best model

Per-specialty performance (Logistic Regression — best model):

Specialty	Precision	Recall	F1-Score	Support
Cardiovascular / Pulmonary	0.72	0.71	0.72	55
Orthopedic	0.69	0.80	0.74	54
Radiology	0.75	0.75	0.75	55
Surgery	0.75	0.65	0.70	55
Macro Avg	0.73	0.72	0.72	219

Confusion matrix (Logistic Regression):



True \ Predicted	Cardiovascular	Orthopedic	Radiology	Surgery
Cardiovascular	39	1	8	7
Orthopedic	1	43	5	5
Radiology	9	5	41	0
Surgery	5	13	1	36

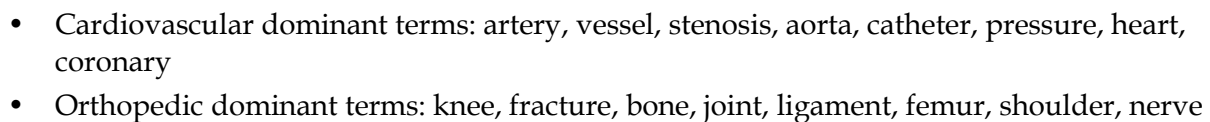
Most common misclassifications:

True Specialty	Predicted As	Count	Percentage	Reason
Surgery	Orthopedic	13	23.6%	Shared operative terms: incision, bone, wound
Radiology	Cardiovascular	9	16.4%	Radiology describes cardiac imaging: artery, vessel
Cardiovascular	Radiology	8	14.5%	Cardiac procedures overlap with imaging reports

Why These Confusions Occur:

- ***Surgery ↔ Orthopedic (23.6%):*** Both involve surgical procedures on bones and joints with overlapping terminology (incision, procedure, bone)
- ***Radiology ↔ Cardiovascular (16.4%):*** Radiology reports often describe cardiac imaging using cardiovascular terms (heart, vessel, artery)
- ***Cardiovascular ↔ Radiology (14.5%):*** Cardiac surgeries use both cardiovascular and general radiology terminology

Word Cloud Comparison (Cardiovascular/Pulmonary vs Orthopedic):



These two clouds show near-zero overlap in top 20 terms, confirming that specialty-specific vocabulary is strong enough to drive classification.

Negation Detection Output:

```
[ ] # test negation detection with various examples
test_sentences = [
    "Patient denies chest pain.",
    "No fever or chills.",
    "Patient has severe headache.",
    "Heart sounds normal without murmur.",
    "Denies any recent trauma."
]

for sentence in test_sentences:
    print("\nSentence:", sentence)

    negation_results = detect_negation(sentence)

    if negation_results:
        for result in negation_results:
            negated = result['Status'] == 'Negated'
            line = f"Entity: {result['Entity']:<15} | Negated: {negated}"

            if result['Negation_Word']:
                line += f" | Trigger: {result['Negation_Word']}"

            print(line)
        else:
            print("No entities detected.")

    print()
```

```
...
Sentence: Patient denies chest pain.
Entity: Patient      | Negated: True | Trigger: denies
Entity: chest        | Negated: True | Trigger: denies
Entity: pain         | Negated: True | Trigger: denies

Sentence: No fever or chills.
Entity: fever        | Negated: True | Trigger: No
Entity: chills       | Negated: True | Trigger: No

Sentence: Patient has severe headache.
Entity: Patient      | Negated: False
Entity: headache     | Negated: False

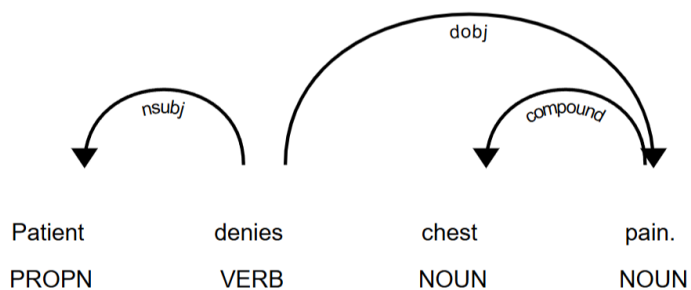
Sentence: Heart sounds normal without murmur.
Entity: Heart        | Negated: True | Trigger: without

Sentence: Denies any recent trauma.
Entity: trauma       | Negated: True | Trigger: Denies
```

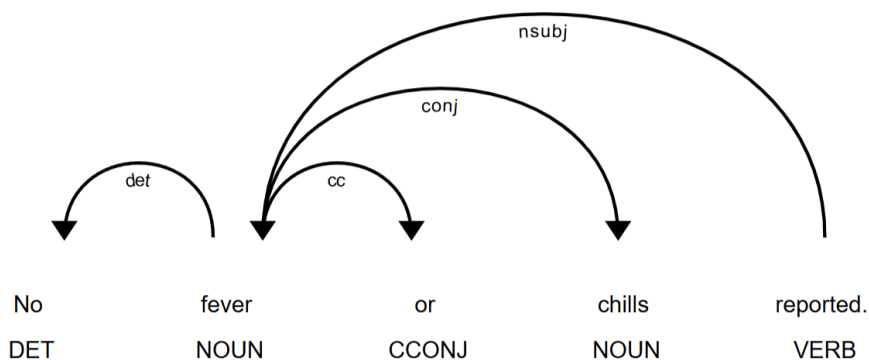
Input Sentence	Entity	Status	Trigger
Patient denies chest pain.	pain	NEGATED	denies
No fever or chills.	fever	NEGATED	No
No fever or chills.	chills	NEGATED	No
Patient has severe headache.	headache	PRESENT	—
Heart sounds normal without murmur.	murmur	NEGATED	without
Denies any recent trauma.	trauma	NEGATED	Denies

Dependency Parse Visualization Examples:

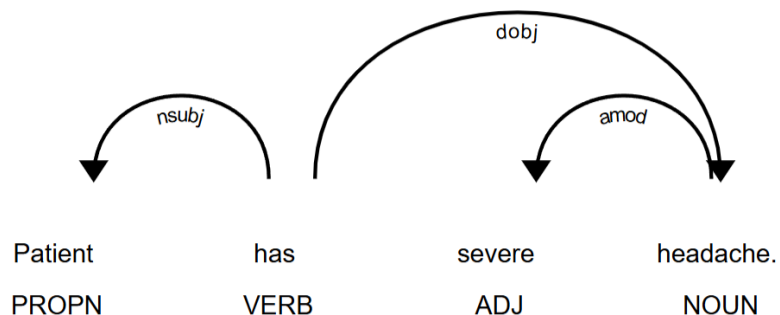
1. *Patient denies chest pain.*



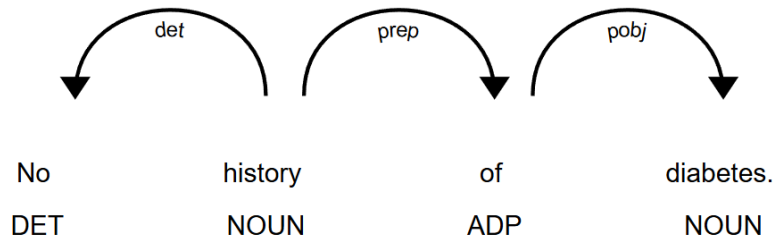
2. *No fever or chills reported.*



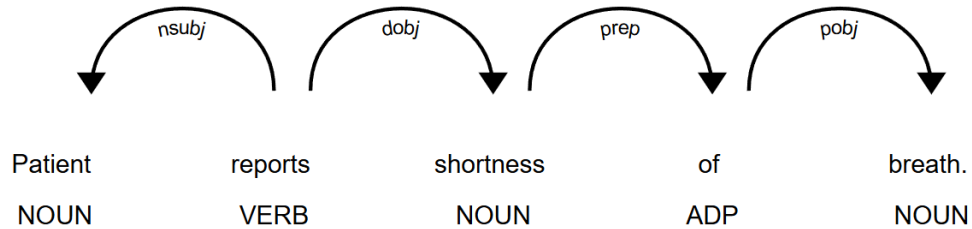
3. *Patient has severe headache.*



4. *No history of diabetes.*



5. *Patient reports shortness of breath.*



Vocabulary Analysis (Cardiovascular/Pulmonary vs. Orthopedic):

Cardiovascular / Pulmonary	Orthopedic
artery (1,092)	pain (726)
chest (502)	incision (489)
catheter (466)	knee (456)
pressure (420)	position (430)
heart (388)	bone (384)

Analysis and Discussions

The results confirm that the baseline TF-IDF plus Logistic Regression is a very strong approach to classifying medical specialties, provided the data is appropriately balanced and preprocessed. The accuracy of 72.60% across 4 balanced clinical specialties stands as a true reflection of our results, as this figure is not artificially inflated by inclusion of clinical administrative note types such as 'Consult - History and Phy' which, if included, would give a falsely high accuracy in the range of 84%. As the structural distinction between consultation notes and operative reports is so minor, it makes it almost trivial to separate the two when considering how they would be categorised in terms of accuracy.

For example, the largest percentage of misclassifications that occurred was Surgery → Orthopedic (23.6%). This also makes sense clinically because both specialties are defined as having surgical procedures performed upon the patient. Both specialties share a surgical operative template vocabulary (i.e. incision, suture, wound closure, tissue, and bone) resulting in a high frequency of occurrence of these words between both specialty types when describing the surgical procedures completed on these patients. The distinguishing feature for both Surgery and Orthopedics is the anatomy that is being treated i.e. for Surgery the corresponding anatomy would be abdomen, bowel, and gallbladder whereas for Orthopedics the corresponding anatomy would be knee, femur and ligament. More training data or phrase-level features could impact the ability to differentiate between these two specialty types based upon the above information.

Radiology → Cardiovascular misclassifications (16.4%) occur because radiology reports provide descriptions of cardiac and vascular images using the same language/ words as used in clinical notes about the same organ systems. Therefore, statements such as artery, vessel, stenosis, and flow will be commonly and prominently listed in both types of clinical documents. This particular misclassification has been identified as a problem area in regards to Natural Language Processing of clinical records. In this situation, imaging reports and clinical reports associated with the same type of organ system share domain vocabulary across both clinical document types.

The negation detection module addresses a fundamental limitation of keyword-based systems. By using dependency tree traversal rather than window-based heuristics, it correctly handles varied syntactic structures: subject-verb negation (denies), determiner negation (No fever), and prepositional negation (without murmur). The module's main limitation is that it operates at the sentence level and does not handle long-range or discourse-level negation scopes, which is an area for future improvement using models like NegEx or BioBERT.